SCALABLE MACHINE LEARNING FOR GLOBAL INTERNET PERFORMANCE TIERING USING OOKLA NETWORK SPEED DATA

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AGENDA

INTRODUCTION

The internet can be considered as one of the most significant developments of the Digital Revolution, as it has become critical to how people access information, communicate, learn, and navigate their daily lives.

The persistent digital divide leads to unequal internet access and network quality across regions and populations.

Traditional analysis methods struggle to efficiently process and interpret large-scale network datasets.

Goal: Identify meaningful internet performance tiers and generate predictive insights using distributed data processing with PySpark.

THE DATASET.

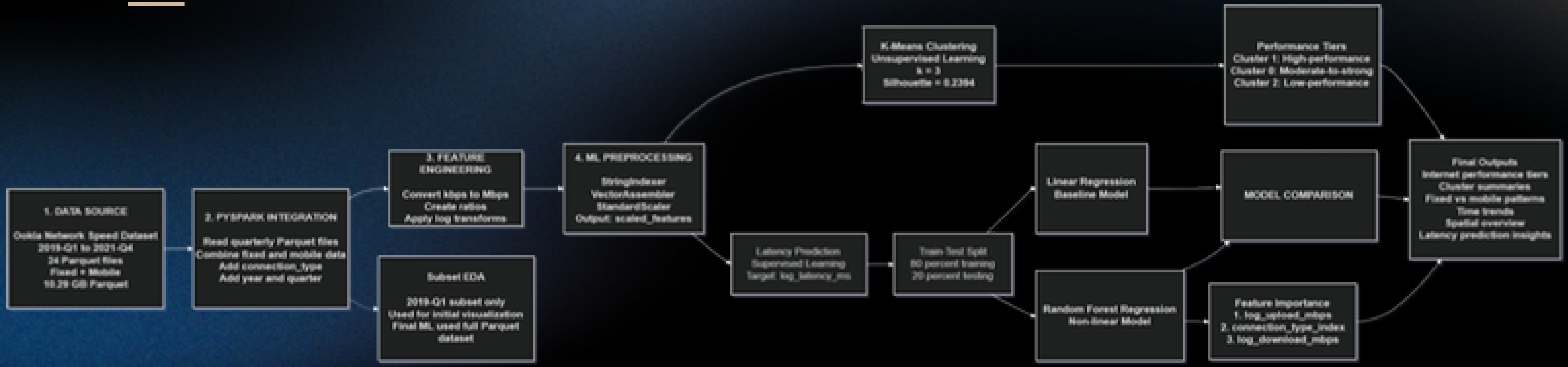
Ookla Open Data serves as our primary dataset.

- Spans from 2019 to 2021
- Original Size: **37.80 GB**
- Parquet Files Pipeline Usage: **10.29 GB**

Selected Key Variables:

- **Latency**
- Download speed
- Upload speed
- Test counts
- Device counts
- Year and quarter
- Connection type

THE MODEL & ALGORITHM.



PySpark

Serves as the primary framework. Implements scalable data ingestion, preprocessing, and machine learning, given the large-scale nature of the Ookla network speed dataset.

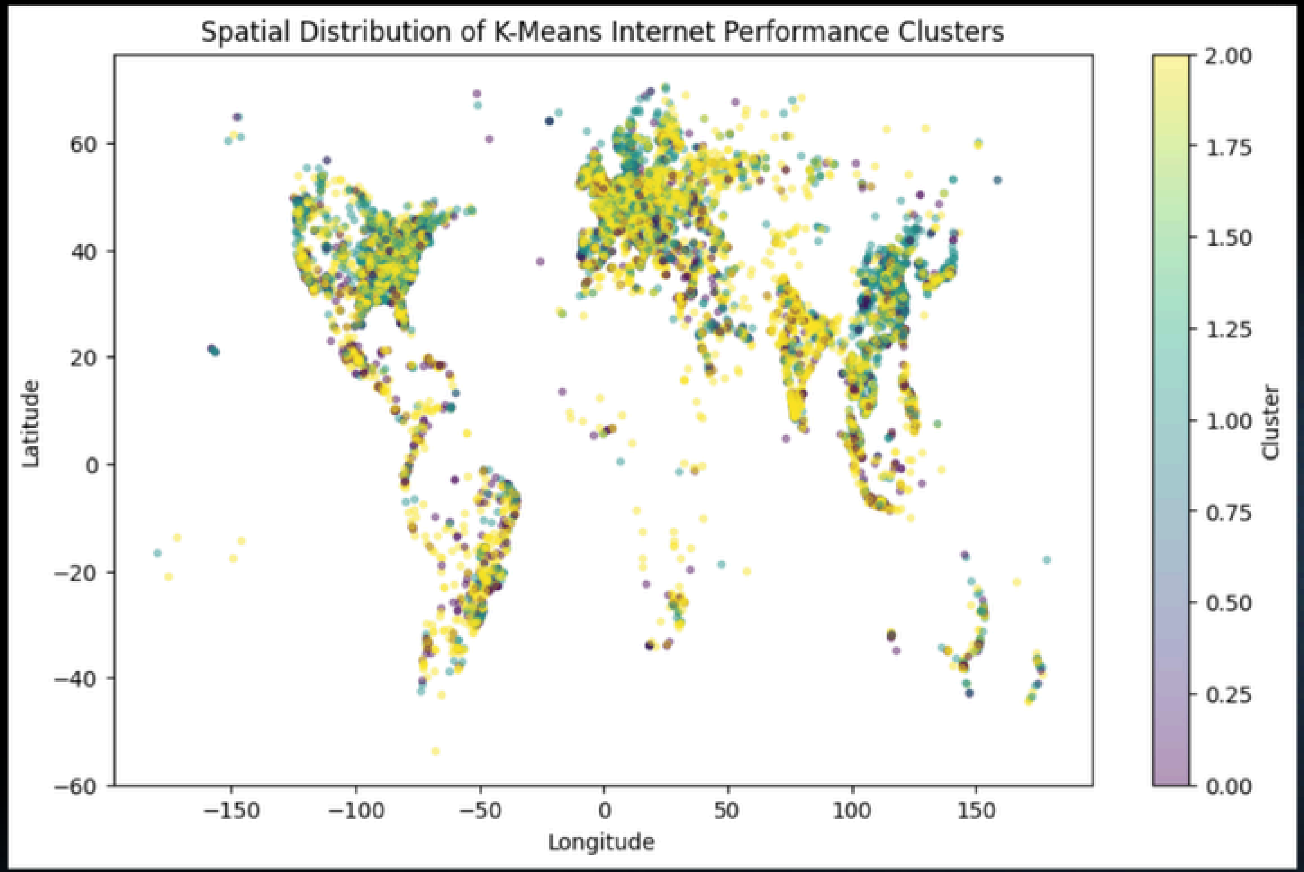
K-Means

Chosen as the Primary Machine Learning Method. The dataset lacks predefined labels for internet performance categories. Its computational efficiency and compatibility with PySpark make it suitable for scalable analysis.

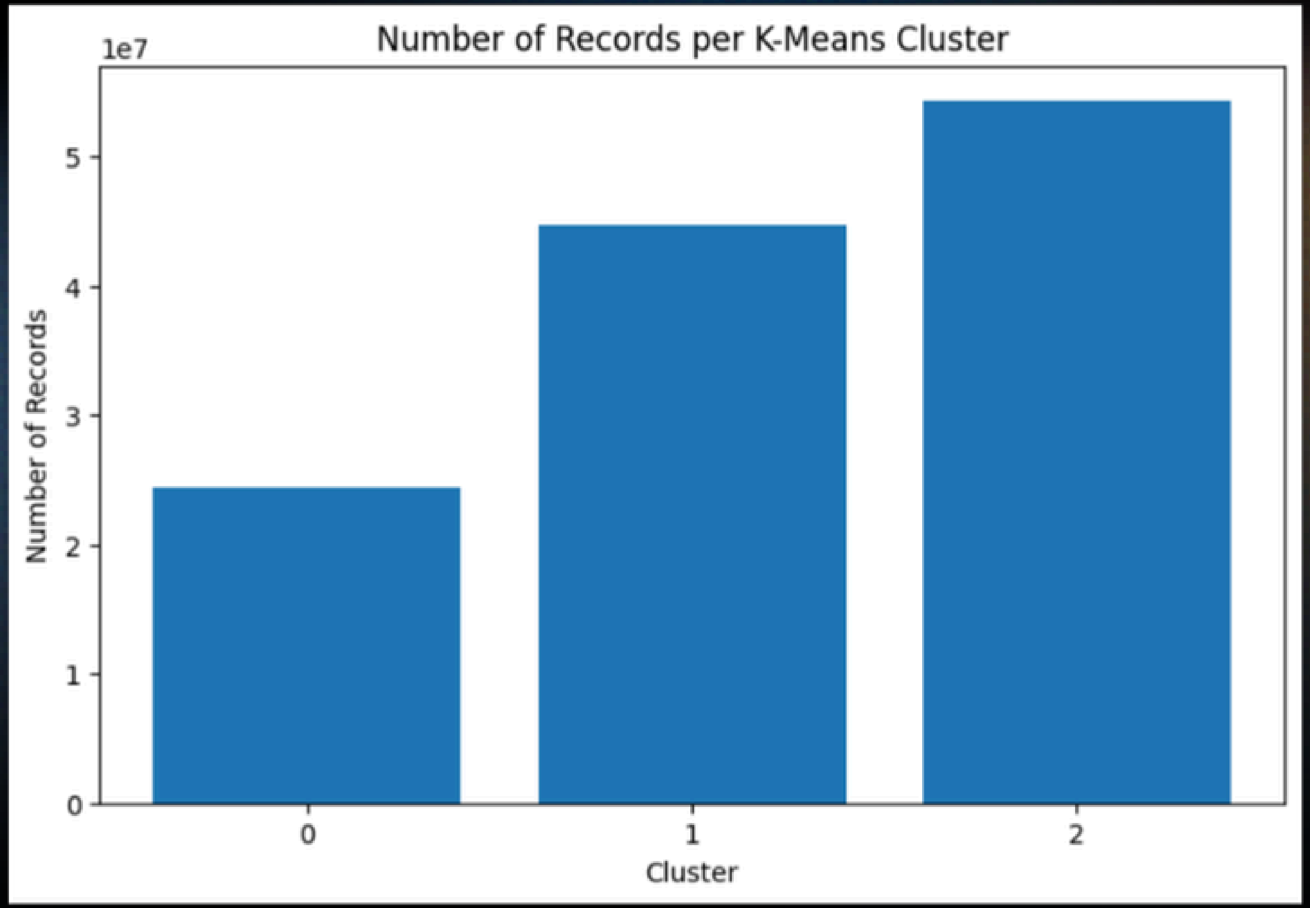
Regression Models

Linear Regression and Random Forest were used. Together, these design choices establish a scalable framework for analyzing and categorizing global internet performance patterns.

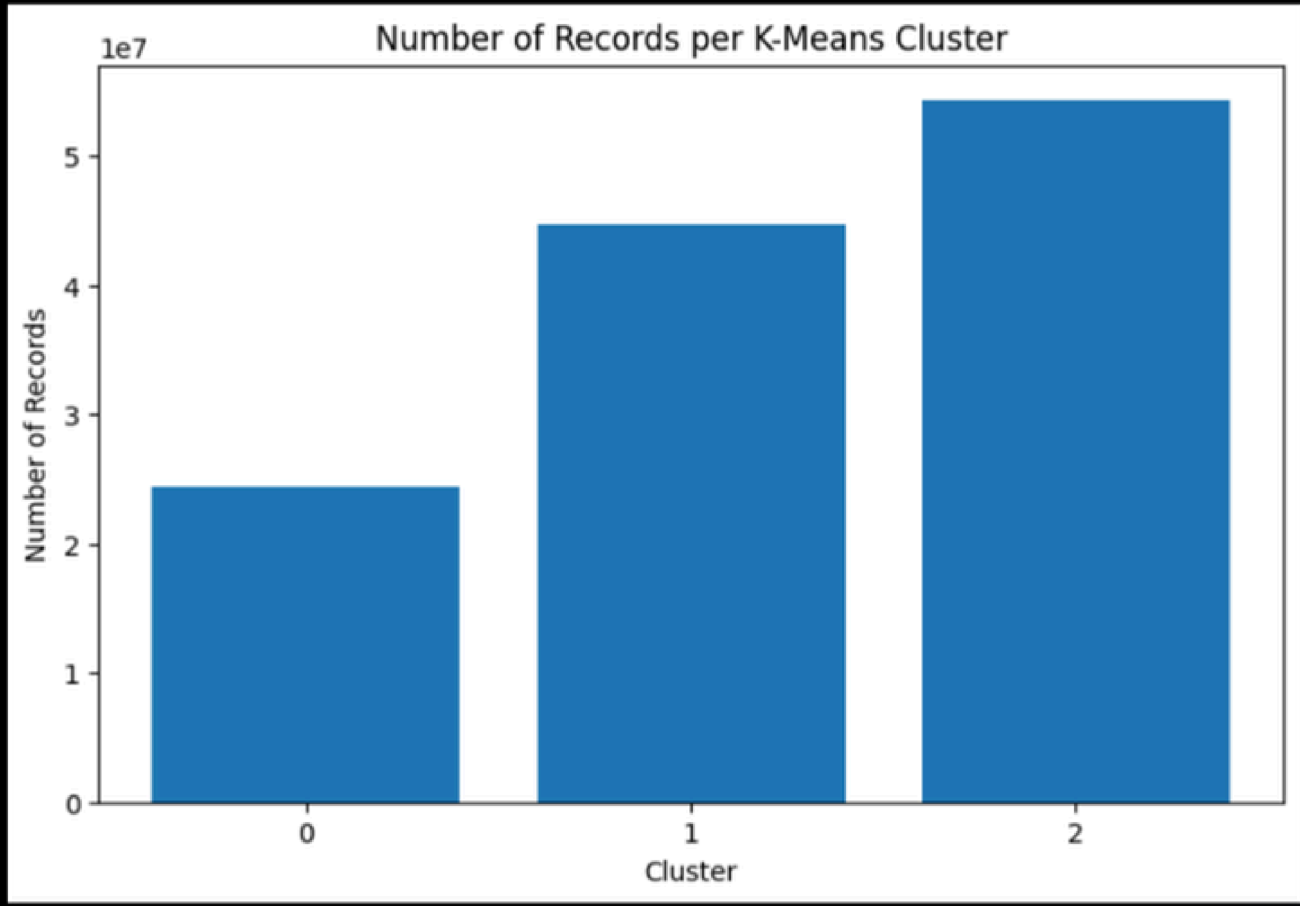
THE RESULTS.

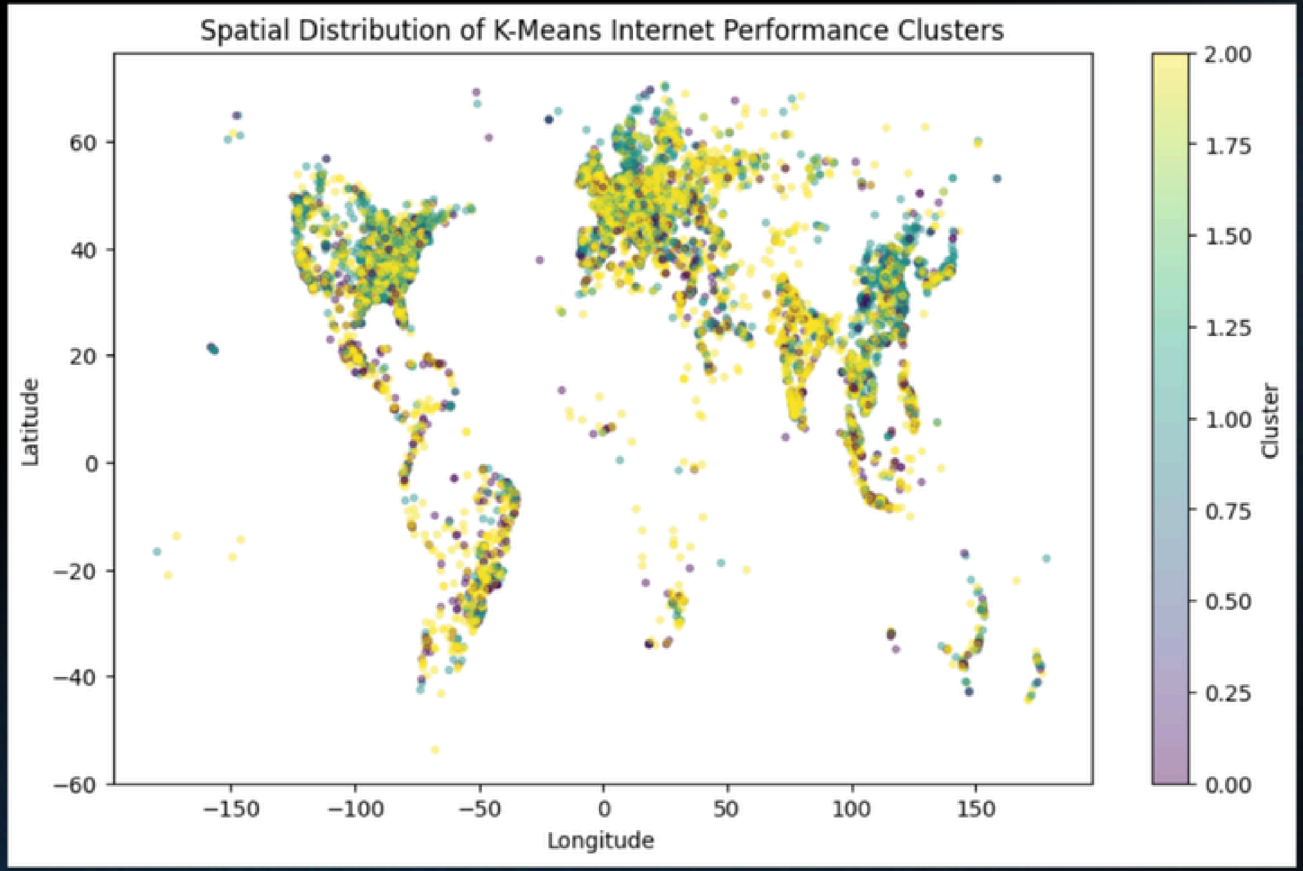


Spatial Distribution of K-Means Internet Performance Clusters
Latitude
Longitude
Cluster
60
40
20
0
−20
−40
−60
−150
−100
−50
0
50
100
150
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00



Number of Records per K-Means Cluster
1e7
Number of Records
Cluster
0
1
2
3
4
5
0
1
2





THE RESULTS.

With a value of **k = 3**, a silhouette score of **0.2394** was attained from the K-Means Clustering Method.

The K = 3 was chosen to create three interpretable clusters

- High-performance
- Moderate/stable-performance
- Low-performance

Clustering was implemented in PySpark MLlib using scalable distributed processing. Although not perfectly separated, this is reasonable because real-world internet quality exists on a continuous spectrum rather than strict categories.

In total, **123,357,711** records were obtained after preprocessing

THE RESULTS.

Linear Regression was used as the baseline interpretable model.

Random Forest Regression was used to capture more complex non-linear relationships between network variables

Random Forest outperformed Linear Regression across all evaluation metrics.

The improvement suggests that latency behavior contains non-linear relationships not fully captured by the linear baseline model.

However, the moderate performance indicates that latency is also influenced by external factors such as routing distance, congestion, and infrastructure quality.

Model	RMSE	MAE	R^2
Linear Regression	0.6745	0.4864	0.3822
Random Forest Regression	0.6550	0.4696	0.4174

DISCUSSION

Scalable ML successfully identified meaningful tiers of internet performance from large-scale Ookla data.

Results are predictive and descriptive, not causal.

The difficult aspects we ran into were the 10.29 GB of Parquet files, which caused runtime, memory, and disk-space limitations; full-data Spark operations and K-Means training were computationally expensive; and the Visualization required sampled or aggregated outputs instead of full-data plotting. Interpreting moderately separated clusters required careful analysis.

Limitations to be stated include:

- Visualization relied on pilot samples due to hardware constraints.
- K-Means clusters showed only moderate separation.
- Regression models used limited network variables and excluded factors such as congestion, routing, and infrastructure quality.
- Local Spark environment constraints affected runtime and implementation decisions.

RECOMMENDATIONS

01 Expanded Data Integration

Future work may improve analysis by incorporating additional geographic and socioeconomic variables. Integrating these factors with Ookla performance data could provide deeper insights into the digital divide and regional connectivity disparities.

02 Improved Clustering Analysis

Future studies should explore additional K-Means configurations by testing different values of k. Comparing clustering performance using the silhouette score, within-cluster sum of squares, and interpretability may produce more refined internet performance tiers.

03 Advanced Predictive Modeling

Future work may improve latency prediction by implementing Gradient Boosted Tree Regression, tuning Random Forest hyperparameters, and applying cross-validation. These approaches could better capture complex latency behavior, although they would require greater computational resources.

THE CONCLUSION.

SMP

SCALABLE ML PIPELINE

This project developed a scalable PySpark-based machine learning pipeline using the Ookla Internet Speed Dataset. The pipeline performed large-scale preprocessing, feature engineering, clustering, and regression analysis on Parquet-based internet performance data.

IPT

INTERNET PERFORMANCE TIERING

K-Means clustering successfully identified three interpretable tiers of internet performance. The resulting clusters showed clear differences in download and upload speeds, latency, and the composition of connection types.

LPI

LATENCY PREDICTION INSIGHTS

Random Forest Regression outperformed Linear Regression across all evaluation metrics, suggesting that latency behavior is influenced by more complex non-linear relationships.

Overall, the study demonstrates how scalable machine learning can generate interpretable insights from large-scale global internet performance datasets.

THANK YOU

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